

LAN Q. LUU

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EDUCATION

Portland State University | Portland, OR

Bachelor of Science in Computer Science | Expected December 2026 | GPA: 3.99

Mt. Hood Community College | Gresham, OR

Associate of Science Transfer in Computer Science | June 2024

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, C, SQL

Frontend: React, Vite, React Router, Bootstrap, HTML/CSS

Backend & Databases: Node.js, Express, Hono, REST APIs, MySQL, PostgreSQL, SQLite

Testing: Jest, Supertest, Vitest, React Testing Library

Tools: Git, GitHub

SOFTWARE DEVELOPMENT EXPERIENCE

IDX Exchange — Software Development Engineer Intern

June 2026 – September 2026

- Built an individual full-stack property search application using React, Node.js, Express, and MySQL, supporting multi-criteria search, server-side pagination, and detailed MLS-style property views.
- Designed REST API endpoints with input validation, parameterized MySQL queries, dynamic filtering, and allowlisted sorting to safely query and serve MLS-style property data.
- Developed 68 passing frontend and backend automated tests using Jest, Supertest, Vitest, and React Testing Library, covering API behavior, validation, UI components, navigation, and error paths.
- Improved frontend reliability with loading/error states, React Error Boundary recovery, defensive data handling, and stale-request protection to prevent outdated search responses from overwriting newer results.

VolunteerNet.org — North Plains Volunteer Network — Software Developer, PSU Capstone

Summer 2026 – Fall 2026

- Extended a React/TypeScript, Hono, and PostgreSQL application to improve administrator request management, enabling coordinators to identify assigned volunteers and access participant information for matched service requests.
- Implemented admin-specific API and PostgreSQL query paths while preserving the application's existing role-based access controls and row-level security model.
- Delivered changes through feature branches and merged GitHub pull requests, integrating frontend, API, and database changes within an established team codebase.

SELECTED PROJECTS

ChocAn Data Center — Team Software Engineering Project

C++, SQLite, Make, Git/GitHub

- Designed and implemented the central application integration layer and core Member, Provider, and Service domain models for a modular healthcare service-management application.
- Developed Manager and Provider terminal workflows, contributed substantially to Operator functionality, and established the initial SQLite integration connecting validation, provider-directory, service, and reporting operations.

String DSL Interpreter — Programming Languages Project

Python, Lark

- Built a tree-walking interpreter for a dynamically typed expression language supporting arithmetic, Boolean logic, conditionals, lexical scoping, recursion, closures, and runtime type checking.
- Designed a Lark grammar and AST transformations for a string-oriented DSL, defining concrete syntax, operator precedence, and string-manipulation operations.

ADDITIONAL EXPERIENCE

Analog Devices — Wafer Operator II | Beaverton, OR

July 2025 – Present

- Operate semiconductor wafer-processing equipment in a high-precision cleanroom environment, maintaining consistent production throughput across standard and value-added manufacturing processes.
- Serve as backup lead during lead absences, supporting workflow coordination, production priorities, and issue escalation to maintain continuous operations.
- Selected to mentor and train a new employee on the ENDR8 toolset while expanding equipment proficiency through cross-training on the ENDR8BARR toolset.